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Managing Opportunistic Consumer Returns in Retail Operations

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Retailers use lenient return policies to stimulate demand and increase revenues, as such policies help customers assess uncertain product valuations at low (or no) return cost and yield higher equilibrium prices. However, generous refunds also yield unintended consequences such as *opportunistic returns*, which take place when customers intentionally "rent" a product for short term use. Accounting for 11% of all product returns in the United States in 2017, opportunistic returns prompt retailers to seek tactics to address adverse revenue and cost implications. We consider two alternative proposals using a price- and refund-setting newsvendor framework with two customer types: *honest returners* and *renters*. The first proposal, *targeted-refunds*, uses retail analytics firms to distinguish renters from honest returners, and implements return policies tailored for each segment. The second proposal, *menu-of-refunds*, presents customers multiple price-refund pairs, and lets them self-select. We compare and contrast the optimal decisions and the profit implications of both proposals with respect to two benchmark settings: one without any renters (Su, 2009), and another proposal, *uniform-refunds*, wherein the retailer merely re-optimizes its decisions while acknowledging that renters exist. We characterize the conditions under which the menu-of-refunds proposal separates customer types, and thus matching *or* exceeding the performance of the targeted-refunds proposal. Furthermore, we study several alternative model specifications to confirm that our main finding concerning the effectiveness of the menu-of-refunds proposal is robust.

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1. Introduction

As noted in the 2015 edition of the Consumer Returns in Retail Industry Annual Survey[1] published by the National Retail Federation (NRF), North American retailers process $260.5 billion worth of returned merchandise annually, which corresponds to more than 8% of total sales. Whereas some returned merchandise can still be resold at premium prices, some others, such as open-box

[1] https://nrf.com/resources/retail-library/consumer-returns-the-retail-industry-2015

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electronics, have to be discounted significantly. Therefore, retailers have an incentive to better understand the underlying reasons why consumers return, and whether the lenient, sometimes "no-questions-asked," return policies are partly to blame. To this end, the aforementioned NRF survey also highlights that about 4 to 6% of returns (estimated between \$9.1 and \$15.9 billion annually) are products that consumers did not have the intention to keep at all; in other words, these consumers opportunistically "rented" products for short-term use knowing that they could be returned for a generous, possibly full, refund. Examples include large screen TV purchases before significant sports events, and apparel items worn once for an event while the tag remained attached. A recent Wall Street Journal (WSJ) article cites the 2017 edition of the NRF survey of 63 retailers in the United States, and reports increases in both of the aforementioned statistics, noting that returns accounted for 11% sales in 2017, and 11% of returns were likely fraudulent.[2] A short-sighted solution for retailers to combat this behavior would be to increase restocking fees; however, in that case, retailers would also risk losing the marketing benefits of lenient return policies, and anger the rest of their customers.

Recognizing the distinct behavior of the aforementioned *renters* versus *honest returners*, where the latter segment of consumers return only if they found themselves unsatisfied with their purchase, retailers have increasingly considered more nuanced strategies than simply increasing restocking fees. One such strategy is to dedicate internal resources or hire third-party data analytics firms such as The Retail Equation (TRE) to track customers' past purchase/return behavior in an attempt to distinguish renters from honest returners, and implement targeted/discriminatory return policies for each segment.[3] Specifically, these software solutions would flag renters during check-out, and communicate to such customers that their sales would be considered final; we will henceforth refer to this proposal as the "targeted-refunds" proposal. Although this proposal looks promising as evidenced by several TRE white papers detailing successful implementations, there are also significant drawbacks as the retailers adopting these software solutions have incurred high costs (e.g., due to licensing, maintenance/technical support, upgrading), and have oftentimes faced backlash from honest returners who may be falsely identified as renters. The aforementioned WSJ article lists prominent retailers such as Best Buy, J.C. Penney, Home Depot, Sephora, and Victoria's Secret among TRE's clients, while providing numerous accounts of customer complaints warning retailers of long-term customer retention concerns.

[2] "How Your Returns are Used Against You at Best Buy, Other Retailers." *Wall Street Journal.* March 13, 2018.
[3] https://www.theretailequation.com/frequently-asked-questions/

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Another recent trend for online retailers is to decouple physical products from after-sales services, and give their customers the option to bundle these services at an additional cost. Jet.com, which was acquired by Walmart in 2016 for $3.3 billion, is one prominent example, and offers its customers several purchasing options at different price points depending on what service options a customer chooses to bundle. For instance, as noted in Mohammad (2015), relative to what is referred to as the "starting price," a customer can enjoy discounts for paying by a debit card instead of a credit card, or by waiving the free returns option and agreeing to pay a restocking fee in case of a product return. Najafi and Duenyas (2019) refer to this latter provision concerning returns as a "conditional markdown," which we investigate in this paper, particularly with respect to its effectiveness relative to the targeted-refunds proposal. More precisely, we study the "menu-of-refunds" proposal, wherein the retailer presents its customers two price-refund pairs, i.e., high-price-high-refund and low-price-low-refund options, and demonstrate that this proposal can mitigate the adverse profit implications of opportunistic consumer returns.

Presuming that the retailers face two distinct customer segments with respect to their purchase and return behavior, i.e., renters versus honest returners, our study investigates the effectiveness of the targeted-refunds and menu-of-refunds proposals by comparing them with each other, as well as relative to two benchmark scenarios: one as in Su (2009) with no renters, and another in which the retailer acknowledges that renters exist, and responds by merely revising its optimal price and refund decisions, a proposal we will henceforth refer to as the "uniform-refunds" proposal.

Our modeling approach entails a newsvendor setting wherein a retailer facing aggregate demand uncertainty makes optimal price, refund, and quantity decisions. Any unsold inventory as well as all returned products are salvaged at a clearance price below procurement cost. We permit two distinct demand segments: (i) renters with known heterogeneous short-term valuations for the product, who are pre-determined to return the product after temporary use, and do not associate any long-term value for keeping the product, (ii) honest returners, who have uncertain but positive long-term valuations of the product, who would return the product only if they realized after their purchase that they are unsatisfied with it. More specifically, the honest returners do not know their own (long-term) valuations *ex-ante*, and only after the purchase, i.e., *ex-post*, they decide whether to keep or return the product.

For the uniform-refunds proposal, we solve the retailer's profit optimization problem subject to the constraints that the price and refund decisions (consistent with utility theory) leave each transacting customer with a non-negative surplus. For the targeted-refunds proposal, we solve

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the retailer's profit optimization problem subject to the aforementioned non-negative utility constraints, while also assuming that a "customer tracking solution" successfully distinguishes renters from honest returners. As such, the retailer applies to each customer transaction a refund option intended for that customer type. Finally, for the menu-of-refunds proposal, the retailer sets two price-refund pairs each intended for one customer type. Yet, this is done without prior identification of customer types, and so each customer would be free to self-select the option, i.e., the high-price-high-refund option or the low-price-low-refund option, which leaves him/her with the highest net surplus. In other words, for the retailer's profit optimization problem to yield a demand segmentation outcome wherein each customer chooses the price-refund pair intended for that customer, the optimal price-refund pairs must satisfy the *individual rationality* and the *incentive compatibility constraints* for each transacting customer.

To assess the relative profitability of each proposal, we first study models that allow optimal refunds, i.e., possibly less than the price yielding non-zero restocking fees. As such we establish findings relative to Su (2009), who found the optimal partial refund to equal salvage value when selling to a customer base with no renters. Our most managerially significant finding is that *a menu-of-refunds proposal with a high-price-high-refund option featuring a refund exceeding the salvage value can properly incentivize customers to carry out purchase (and, possibly return) decisions consistent with (and, therefore revealing) their return behavior, and thus matching or exceeding the profit performance of an optimal targeted-refunds proposal.* We confirm the same when the retailer offers a full refund to one or both customer types, a pricing practice that retailers utilize predominantly, if the fraction of renters in the customer base and the salvage value is below respective thresholds. For high-clearance-price scenarios *or* in markets with a large volume of renters, we find it worthy to adopt targeted-refunds to complement full refunds for honest returners.

In §8, we study several alternative model specifications to verify that our main finding regarding the effectiveness of the menu-of-refunds is robust. Specifically, we extend our formulations in the following aspects: We permit rented (by renters) and returned (by honest returners) products to be resold at different salvage values. We consider the possibility that some renters with high short-term valuations may keep the product they initially intended to return. We study targeted-refunds with noisy identification; in other words, we allow the possibility of errors in distinguishing between renters and honest returners. Finally, we let honest returners to be heterogenous with respect to their known (short-term) or unknown (long-term) valuations. Our primary managerial insight concerning the *the effectiveness of the menu-of-refunds proposal relative to the targeted-refunds proposal* remains intact, and becomes stronger except for the case wherein some renters keep the

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product. Furthermore, our analyses of these alternative model specifications offer limited insights for managing opportunistic returns across multiple periods (as opposed to two periods of full-price sales and clearance) and in the presence of transaction cost of returns.

Our technical findings offer a characterization of the conditions under which the menu-of-refunds proposal effectively mitigates opportunistic consumer returns. In contrast to Su (2009), we find the optimal refund of the uniform-refunds proposal to deviate from the salvage value unless the renters' maximum value-to-rent relative to the honest returners' maximum value-to-purchase is sufficiently low. (We will henceforth refer to this scenario as the "low-rental-revenue-potential" scenario.) In such scenarios, the retailer fully suppresses rental demand by setting a sufficiently high *rental fee*, which we define to be price minus refund. As the renters' maximum value-to-rent increases, we find that the retailer should let the optimal refund to exceed the salvage value if the said salvage value is below a threshold. For such scenarios, we also find that a higher refund permits a higher optimal price, which changes at a lower rate than the price yielding a lower rental fee. In other words, for these "medium-rental-revenue-potential" *and* "low-clearance-price" scenarios, the retailer's optimal response to emerging renters in its customer base is to make renting easier! This is because a higher equilibrium price permits higher margins on *kept sales*, i.e., sales made to honest returners who end up not returning, and a higher equilibrium quantity permits a larger number of transactions at the equilibrium price. Furthermore, in these scenarios, the retailer can use menu-of-refunds to successfully segment customers as effectively as targeted-refunds can by inducing renters and honest returners to self-select the high-price-high-refund and low-price-low-refund options, respectively. In contrast, for "high-rental-revenue-potential" *or* "high-clearance-price" scenarios, we find that the retailer is better-off by lowering the equilibrium refund below the salvage value, thus making it costlier to rent. In these scenarios, menu-of-refunds fails to segment customers, and thus the retailer must resort to targeted-refunds to improve upon uniform-refunds.

The rest of the paper is organized as follows. In §2 we review the relevant literature. We specify our endogenous demand model and present the benchmark model of Su (2009) in §3. We characterize solutions for each of the three proposals under consideration in §4, 5, and 6, respectively. Then, in §7, we compare and contrast the performance of these three proposals, first with optimal refunds, and then with full refunds. In §8, we extend our base model to incorporate several economic factors concerning consumer behavior as well as retailer's operations and costs to highlight that our most significant finding concerning the effectiveness of menu-of-refunds sustains without loss of generality. In this section, we also discuss implications for settings we do not explicitly study in this paper. We conclude the paper and offer managerial guidelines in §9. An electronic companion

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complements the paper in which Appendix A contains technical proofs, and Appendix B provides detailed mathematical analysis supplementing the main findings we present in the main text.

2. Related Literature

The issue of managing returns in retail has received significant interest in operations management, marketing, and economics, where the existing works have focused on how increasingly lenient return policies impact consumers' purchase/return decisions and retail profits. A recent survey by Abdulla et al. (2019) offers a detailed review and classification of this literature. Researchers have specifically investigated the trade-off between reduced consumer risk regarding uncertainty of product fit, which drives sales, versus the increased returns volume, which spikes reverse logistics costs. Along these lines Heiman et al. (2002) portray money-back-guarantees (MBGs) as financial put options while Moorthy and Srinivasan (1995) argue how lenient returns policies increase perceived product quality. Both Davis et al. (1995) and Che (1996) highlight retail profit implications of MBGs, whereas Akçay et al. (2013) focus on the same by integrating resale of open-box items to sales of new products, and both Shulman et al. (2009) and Shulman et al. (2011) emphasize reducing misfit between consumers' tastes and products by providing more information in monopoly and competition settings. Altug (2012) also focuses on balancing benefits and drawbacks of lenient return policies by studying a monopolist retailer selling a single product across multiple periods. Our work also assesses a trade-off associated with utilizing a more lenient return policy for a monopolist, but our focal issue, which is the interplay consumers' enhanced perceived value and opportunistic returns, has not been considered in any of the aforementioned studies.

Whereas the aforementioned papers highlight when pros of lenient returns policies dominate cons, Shang et al. (2017b) focus on the monetary value of a "full refunds" returns policy by using a data set from eBay transactions. They estimate such value to be about 3% of product value, thus cautioning retailers against blindly offering overly generous refunds. Su (2009) also highlight the sub-optimality of full refunds by studying a price- and refund-setting newsvendor setting, and Altug and Aydinliyim (2016) show optimal refunds in the same setting to be even lower (possibly zero) when consumers are strategically discount-seeking. With that being said, Altug and Aydinliyim (2016) also note that consumers' transaction costs, salvage prices exceeding product cost, aggregate demand stimulating effects of lenient return policies, and competition among retailers with asymmetric clearance capabilities can push equilibrium refunds higher, possibly even to the full-refund level. Our study also utilizes the price- and refund-setting newsvendor setting of the aforementioned two studies, but also permits some consumers to demonstrate opportunistic return

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behavior. In addition, we study mechanisms beyond optimal pricing, namely targeted-refunds and menu-of-refunds proposals to mitigate the adverse profit consequences of opportunistic returns. Notably, our equilibrium analysis of the menu-of-refunds proposal provides theoretical support for the empirical work of Shang et al. (2017b) regarding the value of a full-refund option retailers present to consumers, which also finds practical support in case Jet.com's business model and incentive-compatible pricing scheme.

Within the large body of literature that focuses on returns, consumers' opportunistic returns, also referred to as "wardrobing," has received conspicuously scarce attention. In marketing, Hess et al. (1996), Chu et al. (1998), and Davis et al. (1998) were the first works to study this phenomenon using relatively simple demand models where all consumers engage in wardrobing and product fit is captured by a one-for-all and exogenous match probability. In contrast, our model permits some customers, i.e., honest returners, to not engage in opportunistic behavior, and captures endogenous returns volumes for both honest returners and renters as they are influenced by prices and refunds, as well as honest-returners' (ex-post) and renters' (ex-ante) heterogenous valuations. Furthermore, none of the aforementioned studies on wardrobing considers operational issues that arise from demand uncertainty, which, alongside returns is the main driver behind excess inventory a retailer needs to salvage below cost. The price-setting newsvendor model we employ to formulate endogenous and uncertain demand is consistent with earlier works such as Dana and Petruzzi (2001), Su and Zhang (2008), Su (2009), Altug and Aydinliyim (2016). A recent paper by Shang et al. (2017a) is the most similar to ours in terms of how it deviates from and adds onto the earlier studies on wardrobing; however, our study has several notable distinctions: Specifically, we consider demand uncertainty and capture honest returners' and renters' valuations by using general distributions. Furthermore, we analyze settings wherein renters always return the product *as well as* when some renters may keep the product. Finally, we show menu-of-refunds to be as effective a strategy as targeted-refunds to combat opportunistic returns.

Lastly, our analysis of targeted-refunds allows us to relate and contribute to the literature on behavior- and cost-based pricing practices, which highlights the perils of engaging with customers that are too costly to serve. As Narayanan and Kaplan (2001) note, a firm can enjoy up to 250% of the profit it generates by serving "all" of its customers if it can correctly identify and serve "only" the truly profitable customers, a segment that could be as little as 20% of the entire customer base. Shin et al. (2012) employ this logic to study how a firm can profitably discriminate among its customers by targeting each segment with prices it identifies based on past service cost for each segment. We employ a similar idea as we formalize our targeted-refunds proposal, where the

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discrimination materializes via what refund each customer segment is presented with. In a recent working paper, Akturk et al. (2019) also focus on analytics-based technologies to track customers or products to combat fraudulent returns. Our focus is to highlight conditions under which menu-of-refunds can be at least as effective as targeted-refunds, which would require implementation costs and cause customer dissatisfaction.

3. Model Description

We study a monopolist retailer's price, refund, and quantity optimization using a newsvendor framework. We denote price by p, refund by r, and stocking quantity by Q, and when there are distinct alternatives for these decisions, we append appropriate subscripts/superscripts. The retailer faces uncertain aggregate demand D which follows general distribution F. As is standard in newsvendor models, the retailer procures at unit cost c, and sells at price p ($p > c$) while offering refund r ($r \leq p$) for each returned unit. Leftover, i.e., returned and unsold, units are salvaged at salvage value s, where we assume $s < c$ to avoid trivial solutions.

We let consumers obtain utility from using and owning the product in short- and long-term, which we denote by u_S and u_L, respectively. Short-term utility reflects known, i.e., deterministic, valuation of a consumer's first use of the product, whereas long-term utility corresponds to the unknown, i.e., stochastic, value of owning the product and its repeated use over time. The total value a consumer associates with a product is therefore the sum of these short- and long-term valuations. We permit the consumer base to comprise two types of customers, namely, the *honest returners* and *renters*, and assume fraction X (where $0 < X < 1$) of the customers are renters. These customers differ in their short- and long-term valuations. More precisely, honest returners intend to keep the product, yet, they are also uninformed about its true value *prior* to purchase and do not associate with it any short-term utility, i.e., $u_S = 0$. As such, in our base model, each honest customer has valuation $u_H = u_L$ of the product, which follows distribution G_L with support $[0, V_L]$, and consistent with demand models incorporating product returns in extant literature, for example, Su (2009), Altug and Aydinliyim (2016), we assume that an honest returner decides *ex-ante* to purchase the product if and only if $\mathbb{E}\max\{u_L, r\} - p \geq 0$.[4] After the purchase decision the honest returner learns his (customer-specific) true valuation u_L of the product, chooses *ex-post* to keep the product if and only if $u_L - r \geq 0$, and returns it to the retailer otherwise. This keep-or-return decision does not depend on price p, as the paid price becomes a sunk cost for a completed purchase, and yields fraction $G_L(r)$ of honest returners returning the product.

[4] Each honest returner's *ex-post* true valuation u_L is customer-specific. Nevertheless, all u_L values are drawn from the same distribution G_L, yielding the same $\mathbb{E}\max\{u_L, r\} - p$. As a result, all honest returners' *ex-ante* purchase/no-purchase decisions coincide.

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In contrast with honest returners, each renter derives *only* short-term known utility $u_R = u_S$ for using the product, and does not associate positive long-term value with it, i.e., $u_L = 0$. Therefore, if he purchases the product at price p, he derives utility u_S prior to returning it for refund r. Thus, a renter with short-term valuation u_S rents the product if and only if $u_S - (p-r) \geq 0$, and does not participate otherwise. We assume that renters' short-term utility values follows distribution G_S with support $[0, V_S]$, and each renter knows his u_S value; yet, the retailer only knows the distribution. In other words, price p and refund r induces fraction $1-G_S(p-r)$ of renters to acquire the product for short-term use, and then return it. To highlight this short-term use behavior of this customer segment, we will henceforth refer to $p-r$ as the "rental fee." The aforementioned demand dynamics split uncertain aggregate demand D into four segments for a given p and r:

(i) The *kept sales* segment, which includes honest returners who buy and keep, representing fraction $(1-X)\,(1-G_L(r))$ of the demand. Sales to this segment would yield margin $p-c$, which we can re-write as $(p-r)+(r-s)-(c-s)$.

(ii) The *return* segment, which includes honest returners who buy and return, representing fraction $(1-X)G_L(r)$. Transactions with this segment would yield margin $(p-r)-(c-s)$.

(iii) The *rent* segment representing fraction $X\,(1-G_S(p-r))$; transactions with this segment would yield margin $(p-r)-(c-s)$.

(iv) The *no transaction* segment, which includes renters who do not transact with the retailer representing fraction $XG_S(p-r)$, thus yielding leftover inventory with negative margin $-(c-s)$.

We denote by $N(p\,,r)$ the total fraction of customers that transact with the retailer given price p and r, which we will henceforth refer to as *fraction of transacting customers*. Then, using $N(p\,,r) = 1-XG_S(p-r)$, we can express the total volume of customers the retailer transacts with as $DN(p\,,r)$ if the retailer has stocked sufficiently, i.e., $DN(p\,,r) < Q$, and Q otherwise. In the latter case of a stock-out, we assume that the retailer's scarce inventory is split proportionally to segments (i)-(iii) described above, yielding $\mathbb{E}\min\{Q\,,DN(p\,,r)\}$ as the *expected transaction volume* at price p. We denote by $\mathbb{E}\min\left\{\frac{Q}{N(p\,,r)}, D\right\}$ the *transaction volume scaled to full demand* (or, in short, *scaled transaction volume*) at price p, and write the retailer's expected profit as

$$\Big((p-r)\big(1-XG_S(p-r)\big)+(r-s)\big(1-X\big)\big(1-G_L(r)\big)\Big)\mathbb{E}\min\left\{\frac{Q}{N(p\,,r)}, D\right\}-(c-s)Q. \tag{1}$$

Expression (1) highlights that the retailer incurs cost c on the entire stocking quantity, and makes at least salvage revenue s on the same, thus a incurring net cost of $(c-s)$ on each procured unit. The retailer secures at least $p-r$ on each transaction at price p including those that yield returns, i.e., $(p-r)$ applied to the expected transaction volume. Finally, the retailer earns additional $r-s$ net of salvage value from all customers in the keep segment. As we will show later, the relative magnitude of r and s will prove to be a significant factor in retailer's optimal decisions as well

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as which of the three proposals under consideration would be most effective. Su (2009) studied a special case of our model with no renters (i.e., $X = 0$), where the retailer solves

$$\max_{Q,r} \Big\{ \big((p-s)-(r-s)G_L(r)\big)\mathbb{E}\min\{Q,D\} - (c-s)Q \Big\} \tag{2}$$

subject to $\mathbb{E}\max\{u_L, r\} - p \geq 0$. Using superscript "b" (which stands for "benchmark") to denote the optimal decisions for this problem, we have

$$r^b = s\,, \;\; p^b = V_L - \int_s^{V_L} G_L(u)du\,, \;\; 1 - F(Q^b) = \frac{c-s}{p^b - s}.$$

In the next three sections, we consider the general case of our model with $0 < X < 1$, and evaluate the merits of three proposals retailers use to mitigate the adverse profit consequences opportunistic returns. We use superscripts *"u"*, *"t"*, and *"m"*, respectively, to distinguish **u**niform-refunds, **t**argeted-refunds, and **m**enu-of-refunds proposals. (We will also use the same superscripts to distinguish decisions made under each proposal, as well as functions of these decisions such as fraction N of transacting customers.) For each proposal, we characterize the retailer's optimal decisions, and compare such decisions and the ensuing retail profit to those for the benchmark solution of Su (2009) as well as the other two proposals. Throughout the paper, we assume that distributions $G_L(\cdot)$ and $G_S(\cdot)$ of customers' valuations are *stochastically increasing*, and satisfy the *increasing generalized failure rate* (IGFR) property. We denote first derivatives by G_L' and G_S', respectively.

4. The Uniform-Refunds Proposal

The first proposal we analyze is the simplest solution a retailer may consider to combat opportunistic consumer returns, which is to revise its optimal decisions to account for the information that fraction X of its consumers are renters. As this proposal prescribes the same price and refund decisions for all consumers, we refer to it as the "uniform-refunds" proposal. We define

$$M^u(p,r) \triangleq (p-r)\,(1 - XG_S(p-r)) + (r-s)(1-X)\,(1-G_L(r))\,.$$

as the retailer's marginal net revenue from transactions prior to clearance sales. In this case, the retailer solves

$$\max_{Q,r,p} \left\{ M^u(p\,,r)\,\mathbb{E}\min\left\{\frac{Q}{N(p\,,r)}, D\right\} - (c-s)Q \right\} \tag{3}$$

subject to $\mathbb{E}\max\{u_L, r\} - p \geq 0$ and $0 \leq p - r \leq V_S$. Note that when the first of the two aforementioned constraints binds, it suffices to optimize only Q and r as price and refund decisions would then be paired. However, in contrast to the benchmark case with no renters, the retailer may optimally choose to not extract all surplus from honest returners, i.e., the constraint $\mathbb{E}\max\{u_L, r\} - p \geq 0$ for problem (3) may not bind. The demand dynamics that underlie this observation are non-trivial, and sustain only when the salvage value s is (sometimes unrealistically) high, and/or there

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is a prohibitively high fraction of renters.[5] As such we will limit our presentation within the main body of this paper to the case where the said constraint binds, and extend our analysis for when honest returners enjoy positive surplus in section **??** within Appendix B.

PROPOSITION 1. *We denote by $IU(Q,p,r)$ the inventory utilization, i.e., the proportion of the expected transaction volume to stocking quantity, and define it as*

$$IU(Q,p,r) \triangleq \frac{\mathbb{E}\min\{Q, DN(p,r)\}}{Q}. \tag{4}$$

Assuming $xG_S^{'}(x)$ is decreasing in x, optimal decisions $Q^u \triangleq Q(r^u)$ and $p^u \triangleq p(r^u)$ of a retailer implementing uniform-refunds simultaneously satisfy

$$1 - F\left(\frac{Q^u}{N(p^u, r^u)}\right) = \frac{(c-s)N(p^u, r^u)}{M^u(p^u, r^u)}, \tag{5a}$$

$$p^u = V_L - \int_{r^u}^{V_L} G_L(u)du. \tag{5b}$$

Furthermore, denoting by $\hat{r}$ the value of r satisfying

$$\frac{\partial M^u}{\partial N}\big(p(\hat{r}), \hat{r}\big)\, IU\big(Q(\hat{r}), p(\hat{r}), (\hat{r})\big) = c - s, \tag{5c}$$

there exists a threshold value V_S^{LB} such that

$$r^u = \begin{cases} \max\{\hat{r}, 0\} & \text{if } V_S > V_S^{LB}, \\ r^b & \text{otherwise.} \end{cases} \tag{5d}$$

We derive (5a) by using the first order condition (FOC) of profit (3) with respect to Q, whereas (5b) ensures the retailer's price and refund decisions extract all consumer surplus from honest returners. Equation (5c) is more nuanced, which we derive by assessing the FOC of profit (3) with respect to r around $Q = Q^u$ and $p = p^u$. To appreciate the intuition that underlies (5c), we state the following lemma.

LEMMA 1. *We have $0 < \frac{dp^u(r)}{dr} \leq 1$, which further implies $\frac{d[p^u(r)-r]}{dr} < 0$, and $\frac{\partial Q^u(r)}{\partial r} > 0$ and.*

Lemma 1 highlights that a more generous refund (i.e., larger value of r) results in a higher equilibrium price. However, as price increases at a slower rate than refund, rental fee $p - r$ decreases. Consequently, return volumes for each customer type increase, i.e., honest returns increase due to a higher refund, and rentals increase due to a lower rental fee. Despite these adverse consequences, it is optimal for the retailer to increase refund until (5c) in Proposition 1 is satisfied, as the retailer also stocks a higher quantity yielding two additional revenue opportunities: The increase in rental

[5] One sufficient condition to ensure $p = \mathbb{E}\max\{u_L, r\}$ holds at equilibrium is $V_S G_S^{'}(V_S) < (1-X)/X$. For example, if G_S follows uniform or symmetric triangular distributions, the said condition reduces to $X < 0.5$, i.e., there are fewer renters than honest returners. Note that the 2017 edition of the annual NRF survey places X at 11%.

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demand satisfied using higher quantity may overcome the reduction in rental fee, and the higher equilibrium price increases the margin on kept sales to honest returners.

The aforementioned coupling between the equilibrium refund and stocking quantity permits the following interpretation of (5c) in Proposition 1: Each additional unit the retailer procures costs $c-s$ net of salvage revenue, the right-hand side (RHS) of (5c), and this additional unit yields a sales transaction at price p with probability $IU(Q,p,r)$, i.e., the second term on the left-hand side (LHS) of (5c), yielding the marginal revenue (net of refund in case of a return) denoted by first term on the LHS of (5c). As such, (5c) characterizes the equilibrium outcome where the *expected* marginal revenue of a procured unit at the equilibrium price and refund equals its marginal cost.

Finally, expression (5d) in Proposition 1 highlights that the equilibrium implied by the aforementioned three conditions sustains only when the equilibrium rental fee induces positive rental demand, i.e., $p^u - r^u < V_S$. Otherwise, with no rental transactions, the benchmark solution of Su (2009) with $r^b = s$ sustains. A relevant managerial question then concerns how r^u when there is rental demand compares to $r^b = s$ for the benchmark case with no renters, which we address in the next:

PROPOSITION 2. *We have $r^u \geq r^b = s$ if-and-only-if*

$$IU(Q^b,p^b,r^b)\left((p^b-s)-\frac{1-G_S(p^b-s)}{G_S'(p^b-s)}\right) > c-s. \tag{6}$$

Note that when all the requisite substitutions are made, condition (6) in Proposition 2 solely relies on problem parameters c, s, and distributions $G_S(\cdot)$, $G_L(\cdot)$, and $F(\cdot)$. As such, it offers a full characterization regarding when the retailer should induce more transactions that yield returns (or, restrict them,) relative to the benchmark scenario with no renters in the consumer base. In a nutshell, using Lemma 1, the retailer lets r^u to exceed s, and increases its stocking quantity accordingly if the marginal cost $c-s$ of procuring that additional unit remains below the *expected* marginal revenue the retailer anticipates from rental/return transaction that additional unit would trigger in the benchmark case. The latter *expected* marginal revenue equals the marginal revenue $(p^b-s)-\frac{1-G_S(p^b-s)}{G_S'(p^b-s)}$ per such transaction multiplied by the probability $IU(Q^b,p^b,r^b)$ that the additional procured unit would trigger such a transaction. The condition in Proposition 2 also permits an interpretation in terms of critical problem parameters s, V_S, and V_L, which we use to formalize thresholds on V_S/V_L in Proposition 3.[6]

[6] We illustrate the profit and optimal proposal implications of these dynamics as well as the influence of salvage value s on thresholds V_S^{LB} and V_S^{UB} in panels (a) and (b), respectively, of Figure 1 in Section 7.

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PROPOSITION 3. *For a given V_L, there exists thresholds V_S^{LB} and V_S^{UB}, which depend on salvage value s, such that:*

(i) We have $r^u = s$ for $V_S < V_S^{LB}$ while no renters transact.

(ii) We have $r^u > s$ for $V_S^{LB} < V_S < V_S^{UB}$.

(iii) We have $r^u < s$ for $V_S > V_S^{UB}$.

To appreciate Proposition 3, we provide a detailed description of the underlying dynamics in section **??** within Appendix B, and here provide a summary. We first recall that the retailer's optimal decisions yield margin $p - r$ on all transactions (i.e., kept sales, returns, rents) as well as additional margin $r - s$ *only* on kept sales against leftover cost $c - s$ on all procured units. Therefore, deviating from $r^u = r^b = s$ hinges upon the retailer's ability to extract maximum margin from sales while optimizing the transaction volume. Specifically, when the rental value-potential is extremely limited, i.e., low V_S/V_L as in Proposition 3(i), the retailer sets $p^u - r^u = V_S$ to price out renters, and $r^u = r^b = s$ to obtain the same margin on kept sales and returns, i.e., $p^u - s$. In contrast, the retailer capitalizes on renters' high willingness-to-pay in high-rental-value-potential scenarios as noted in Proposition 3(iii), and sets a high rental fee by reducing r^u below s. In effect, the retailer makes a smaller margin on kept sales than on rents and returns as $p^u - s < p^u - r^u$. For medium-rental-value-potential scenarios as in Proposition 3(ii), the retailer's decision is even more nuanced and depends also on salvage value s. In low-clearance-price scenarios, i.e., low s, the retailer counters high leftover cost $c - s$ by setting $r^u > s$. As such the retailer extracts a higher margin on kept sales than on rents and returns, i.e., $p^u - s > p^u - r$, while stimulating a higher transaction volume relative to when $r^u = r^b = s$. In contrast, for high-clearance-price scenarios, the retailer sets $r^u < s$ to ensure higher margins on rents and returns than kept sales.

A significant drawback of the uniform-refunds proposal is that the unilateral changes in its policy components help and hurt retail profits concurrently. For example, an increase in price help reduce renting behavior, (i.e., an increase in p implies an increase in rental fee $p - r$,) while, at the same time, reducing honest returners expected net utility, i.e., $\mathbb{E}\max\{u_L, r\} - p$ decreases. In contrast, the retailer may attempt to increase the aforementioned net utility by increasing r, but such a refund adjustment may encourage renting behaviour by reducing the rental fee. One way to overcome this issue is to identify each customer type prior to sales, and discriminate these customer segments by offering them different refunds; next, we analyze one such proposal.

5. The Targeted-Refunds Proposal

In this proposal, we consider the optimal decisions for a retailer which has already dedicated internal resources or hired a third-party data analytics firm such as The Retail Equation (TRE)

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to distinguish renters from honest returners, and implement discriminatory return policies. More specifically, while the retailer makes universal price and quantity decisions, this "targeted-refunds" proposal charges each customer segment a different restocking fee. We use r_R^t and r_H^t to denote the equilibrium refunds the retailer offers to the renters and honest returners, respectively. A retailer implementing this proposal will identify a renter with probability X $(0<X<1)$ and solve

$$\left\{ \begin{array}{cc} \max\limits_{p-r} & \Big((p-r)\big(1-G_S(p-r)\big)\Big)\mathbb{E}\min\big\{Q\,,D\big\}-\big(c-s\big)Q \\ \text{subject to} & 0\le p-r\le V_S \end{array} \right\} \text{ with probability } X. \tag{7}$$

If, instead, an honest customer is identified with probability $1-X$, then the retailer solves

$$\left\{ \begin{array}{cc} \max\limits_{r} & \Big((p-s)-(r-s)G_L(r)\Big)\mathbb{E}\min\big\{Q\,,D\big\}-\big(c-s\big)Q \\ \text{subject to} & \mathbb{E}\{u_L,r\}-p=0 \end{array} \right\} \text{ with probability } 1-X. \tag{8}$$

A significant assumption in the aforementioned formulations is the retailer's ability to correctly distinguish a renter from an honest returner, which, in practice may not be the case, simply because any implementation algorithm would be prone to errors. (For example, most prominent software implementations flag renters when the predicted probability of a customer being a renter exceeds a threshold that is less than 1.) As such, the retail profits obtained by using program formulations (7) and (8) would be upper bounds for any proposal we assess in this paper.

Note that program (7) would be solved if the retailer identified a renter, and thus it entails finding the optimum rental fee $p^t-r_R^t$. Similarly, program (8) would be solved if an honest returner were correctly identified, in which case, the retailer's optimization reduces to the benchmark problem (2) of Su (2009). (Note that, in this case, only refund optimization suffices as the constraint pairs price and refund at equilibrium.) As such we expect the optimal price p^t and honest returners' optimal refund r_H^t to coincide with p^b and r^b, respectively. The optimal order quantity Q^t, on the other hand, would be distinct from Q^b, as the retailer would set optimal decisions prior to the uncertainty regarding consumer types is resolved. In other words, denoting by

$$M^t(p,r_R,r_H)\triangleq X\big(p-r_R\big)\big(1-G_S(p-r_R)\big)+\big(1-X\big)\big((p-s)-(r_H-s)G_L(r_H)\big)$$

the retailer's marginal net revenue, the optimization involves solving, *apriori*, the following:

$$\left\{ \begin{array}{cc} \max\limits_{Q,r_R,r_H,p} & M^t(p,r_R,r_H)\mathbb{E}\min\left\{\dfrac{Q}{N(p\,,r_R)},D\right\}-\big(c-s\big)Q \\ \text{subject to} & 0\le p-r_R\le V_S \\ & \mathbb{E}\{u_L,r_H\}-p=0 \end{array} \right\}. \tag{9}$$

The next proposition offers a full characterization of the retailer's optimal targeted-refunds policy.

Proposition 4. *The retailer's optimal targeted-refunds proposal decisions Q^t, p^t, r_R^t and r_H^t simultaneously satisfy*

$$1-F\left(\frac{Q^t}{N(p^t\,,r_R^t)}\right)=\frac{(c-s)\,N(p^t\,,r_R^t)}{M^t(p^t,r_R^t,r_H^t)} \tag{10a}$$

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$$p^t = p^b = V_L - \int_s^{V_L} G_L(u)du \tag{10b}$$

$$r_H^t = r^b = s \tag{10c}$$

$$IU(Q^t, p^t, r_R^t)\left(p^t - r_R^t - \frac{1 - G_S(p^t - r_R^t)}{G_S'(p^t - r_R^t)}\right) = c - s \text{ if } p^t - r_R^t < V_S. \tag{10d}$$

Furthermore, we have $r_R^t \geq r^b = s$ *if-and-only-if* $r^u \geq r^b = s$.

Proposition 4 highlights an obvious benefit of distinguishing customer types; the retailer can set price and refund for honest returners as accurately as it could when there are no renters, i.e., $r_H^t = r^b = s$. It also highlights that correct identification of renters does not necessarily constrain the refund below the benchmark level, i.e., for several realistic scenarios, $r_R^t \geq r^b = s$. The rationale behind this finding is rather non-intuitive, which we describe next.

Note that when the retailer correctly identifies honest returners, it can limit its losses on honest returners' returned items $r_H + c - s$ by setting $r_H^t = s$. As a result, in scenarios with sufficiently low V_S/V_L the retailer can afford to lose an additional $r_R^t - r_H^t$ on items renters are determined to return, because a large r_R^t also permits a Q^t, and thus higher kept sales. In such scenarios relatively low V_S helps maintain a low rental volume, whereas a relatively high V_L support the price premium and a low volume of honest returns. It is of note for these scenarios that the retailer makes renting even easier than it is under the uniform-refunds proposal, i.e, $p_R^t - r_R^t < p^u - r^u$, as the ensuing price premiums on kept sales justify doing so.

In contrast to the dynamics we described in the previous paragraph, the retailer suppresses the renters' refund offering below the benchmark value, i.e., $r_R^t < r^b = s$, when V_S/V_L is sufficiently high. In such scenarios, equilibrium p^t and Q^t values would also be lower than they would be in benchmark scenario or under the uniform-refunds proposal. Evidently, without the conditions that support a price premium, it is best for the retailer to limit its leftover losses by stocking less. Furthermore, under these scenarios, the retailer should discourage renting by setting the rental fee at a higher level than it is under the uniform-refund proposals, i.e, $p_R^t - r_R^t > p^u - r^u$.

6. The Menu-of-Refunds Proposal

In this proposal we consider the optimal decisions for a retailer which cannot identify customer types, but offers a menu of prices and refunds with the goal of inducing each consumer to self-select the price-refund pair intended for that consumer type. We will refer to this proposal as the "menu-of-refunds" proposal, denote its optimal price-refund pairs by (p_H^m, r_H^m) and (p_R^m, r_R^m), and its optimal stocking quantity by Q^m. When facing this proposal, each honest consumer self-selects the menu offering maximizing $\mathbb{E}\max\{u_L, r_i\} - p_i$ for $i \in \{H, R\}$, whereas a renter with short-term valuation u_S will self-select to purchase by paying p_i and return to receive refund r_i, where

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$i = \arg\min_i\{p_i - r_i\}$ for $i \in \{H, R\}$, if-and-only-if $\min\{p_i - r_i\} < u_S$. To express the optimization problem a retailer implementing this proposal, we define net marginal revenue as in the following

$$M^m(p_R, r_R, p_R, r_H) \triangleq X\big(p_R - r_R\big)\big(1 - G_S(p_R - r_R)\big) + \big(1 - X\big)\big((p_H - s) - (r_H - s)G_L(r_H)\big),$$

and express the retailer's optimization as

$$\left\{ \begin{array}{cc} \max\limits_{Q,(p_H,r_H),(p_R,r_R)} & M^m(r_R, r_H, p_R, p_H)\,\mathbb{E}\min\left\{\dfrac{Q}{N(p_R\,, r_R)}, D\right\} - (c-s)Q \\ \text{subject to} & \mathbb{E}\max\{u_L, r_H\} - p_H \geq \max\left\{\mathbb{E}\max\{u_L, r_R\} - p_R, 0\right\} \\ & p_R - r_R \leq \min\{p_H - r_H, V_S\} \end{array} \right\}. \tag{11}$$

The first constraint in program (11) ensures the honest returners prefer price-refund pair (p_H, r_H) while enjoying non-negative net utility, whereas the second constraint ensures renters who do not walk away without a purchase self-select a transaction at rental fee $p_R - r_R$. We offer a full characterization of the retailer's optimal menu-of-refunds proposal in the next proposition.

Proposition 5. *The optimal menu-of-refunds proposal decisions simultaneously satisfy*

$$1 - F\left(\frac{Q^m}{N(p_R^m\,, r_R^m)}\right) = \frac{(c-s)\,N(p_R^m\,, r_R^m)}{M^m(r_R^m, r_H^m, p_R^m, p_H^m)}, \tag{12a}$$

where the optimal price-refund pairs (p_H^m, r_H^m) *and* (p_R^m, r_R^m) *satisfy*

$$(p_H^m, r_H^m) = \left\{ \begin{array}{ll} (p^u, r^u) & \textit{if } r^u < r^b = s, \\ (p^t, r_H^t) & \textit{otherwise,} \end{array} \right. \tag{12b}$$

$$p_R^m - r_R^m = \left\{ \begin{array}{ll} p^u - r^u & \textit{if } r^u < r^b = s, \\ p^t - r_R^t & \textit{otherwise.} \end{array} \right. \tag{12c}$$

Proposition 5 highlights that the optimal menu-of-refunds proposal *either* coincides with the optimal uniform-refunds proposal *or* the optimal targeted-refunds proposal. In other words, the retailer can implement menu-of-refunds to segment customers and match targeted-refunds' profit performance in scenarios supporting $r^u > r^b = s$, or equivalently $r_R^t > r_H^t = s$.

7. Proposal Comparisons

In this section, we first integrate the optimal policy characterizations we provided in §§ 4, 5, and 6. Our analysis offers a full characterization of model parameters concerning demand and retail margins under which the menu-of-refunds proposal can segment customers as effectively as the targeted-refunds proposal does. Then, given the widespread use of no-restocking-fee policies in retail, we study how the three proposals fare when the retailer permits full refunds on some (or all) returns, we impose necessary restrictions on each of the corresponding optimization problems. Specifically, for *full-uniform-refunds*, we restrict the refund applied to all returns to be equal to price. For *full-targeted-refunds*, we assume that the retailer permits 100% refund for honest returners, whereas we let the refund to be offered to the renters be optimized. Finally, for *full-menu-of-refunds*, we set the refund for the high-price option at 100% of the price, while restricting the refund of the low-price option to 0.

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7.1. Proposal comparisons with optimal refunds

Before formalizing our findings concerning proposal comparisons, we first revisit earlier propositions and note that the conditions under which the menu-of-refunds and targeted-refunds proposal coincide are the same conditions inducing the renter's refund under the targeted-refunds proposal to exceed the salvage value, and thus the honest returners' refund under the same proposal. Mathematically, under such conditions, we have $r_R^t > r_H^t = s$, which further implies that the rental fee for the renters is lower than the same for the honest returners, i.e., $p^t - r_R^t < p^t - r_H^t$. Evidently, for the menu-of-refunds proposal to lure renters to the price-refund pair intended for them, the rental fee of that high-price-high-refund options must be more desirable by the renters, i.e., renters' incentive compatibility constraint $p_R^m - r_R^m \leq p_H^m - r_H^m$ of optimization program (11) must be satisfied. In subsection **??** within Appendix B, we show that, when renters' incentive compatibility constraint does not bind, i.e., $r_R^m > r_H^m = s$, honest returners' incentive compatibility constraint $\mathbb{E}\max\{u_L, r_H\} - p_H \geq \mathbb{E}\max\{u_L, r_R\} - p_R$ holds as a strict inequality. Conversely, $r_R^m < r_H^m = s$ cannot segment renters and honest returners as the retailer intended, and thus, when $r_R^m > r_H^m = s$ is not optimal for the retailer, the menu-of-refunds proposal should mimic the uniform-refunds proposal, i.e., $r_R^m = r_H^m = r^u < s$. We leverage these dynamics as well as Proposition 3, which establishes threshold V_S^{UB} concerning when r^u exceeds or is smaller than s, and formally characterize the retailer's optimal proposal choice at equilibrium as in the following proposition:

PROPOSITION 6. *Denoting by* Π^k *for* $k \in \{u, t, m\}$ *the optimum retail profit when the retailer implements proposal* k, Π^u, Π^t, *and* Π^m *satisfy*

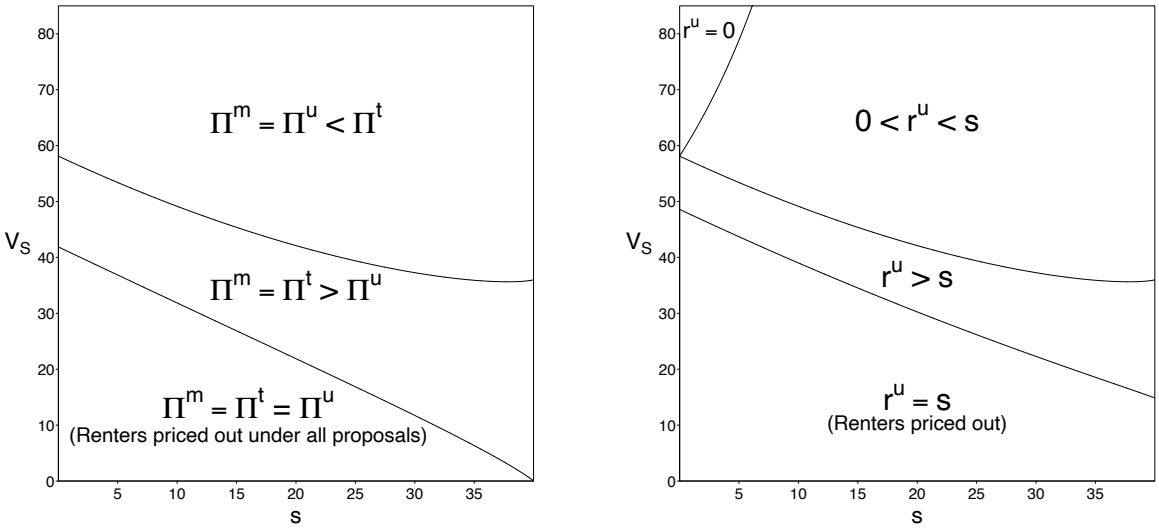
(a) $\Pi^t > \Pi^m = \Pi^u$ *if-and-only-if* $r^u < r^b = s$,

(b) $\Pi^t = \Pi^m > \Pi^u$ *if-and-only-if* $r^u > r^b = s$

(c) $\Pi^t = \Pi^m = \Pi^u$ *otherwise.*

The most significant observation from Proposition 6 and Figure 1 is that the optimal menu-of-refunds proposal can segment customers as effectively as the optimal targeted-refunds proposal can for a wide range of problem scenarios. In subsection **??**, we illustrate that this characterization also applies when X is larger, i.e., scenarios wherein the optimal uniform-refunds proposal does not extract all surplus from honest returners. Figure 1 also illustrates that the menu-of-refunds scenario is effective for retailers selling products with low-rental-revenue-potential (i.e., low V_S) as well as for low-clearance-price products (i.e., low s) in markets with medium-revenue-potential. This begs the question whether companies like Best Buy, which must discount returned products such as open box electronics, should partner at all with retail analytics firms such as TRE, our findings advocate using menu-of-refunds instead.

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(a) Equilibrium profit comparisons **(b)** Equilibrium uniform-refunds proposal

Figure 1 **Scenarios for which targeted-refunds outperforms menu-of-refunds coincide with scenarios supporting $r^u < s$. Parameter values: $V_L = 100$, $c = 40$, $X = 0.2$, and D follows Normal(100,20).**

7.2. Proposal comparisons with full refunds

Given the popular use of "no-restocking-fee" pricing in retail, we now investigate the effectiveness of the three proposals in scenarios wherein at least one customer segment is given the option to return for a full refund. We will use $\hat{p}$, $\hat{r}$, and $\hat{Q}$ with appropriate subscripts and superscripts to differentiate the optimal decisions of the full-refund proposal variations. In our analysis of full-menu-of-refunds, we will also incorporate strict denial of returns for the discounted purchase option. Specifically, we will compare and contrast the menu-of-refunds proposal with a high-price-high-refund option featuring a full refund and a low-price-low-refund option featuring no refund, against a uniform-refunds proposal with $\hat{p}^u = \hat{r}^u$, and a targeted-refunds proposal with $\hat{p}^t = \hat{r}^t_H$. We next state the retailer's optimization problems with the full-refund stipulations. In these formulations, for ease of exposition, we will use p and δ to denote price p_R intended for renters and price differential $p_R - p_H$ between the prices intended for each segment, respectively. We express the retailer's optimal decisions as

$$\hat{p}^u \equiv \arg \max_{\left\{p \,|\, \mathbb{E}\max\{u_L,p\}-p\geq 0\right\}} \Big(\big(1-X\big)\big(p-s\big)\big(1-G_L(p)\big)\Big)\mathbb{E}\min\big\{Q,D\big\} - \big(c-s\big)Q, \tag{13}$$

$$(\hat{p}^t,\hat{r}^t_R) \equiv \arg \max_{\left\{(p,r_R)\,|\,p\leq \min\{\mathbb{E}\max\{u_L,p\},r_R+V_S\}\right\}} M^t(p,r_R,r_H=p)\mathbb{E}\min\Big\{\frac{Q}{1-XG_S(p-r_R)},D\Big\} - \big(c-s\big)Q, \tag{14}$$

$$(\hat{p}^m,\hat{\delta}^m) \equiv \arg \max_{\left\{(p,\delta)\,|\,p-\delta\geq s\,,\,\delta\geq 0\right\}} \begin{cases} \Big(\big(1-X\big)\big(p-s\big)\big(1-G_L(p)\big)\Big)\mathbb{E}\min\big\{Q,D\big\} - \big(c-s\big)Q & \text{if } \delta < \int_0^p G_L(u)du, \\ \Big(\big(1-X\big)\big(p-\delta-s\big)\Big)\mathbb{E}\min\big\{Q,D\big\} - \big(c-s\big)Q & \text{otherwise.} \end{cases} \tag{15}$$

For expression (15), note that the first branch corresponds to the scenario wherein the retailer fails to segment consumers as discount δ is not sufficiently high. In this case, both types of customers

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choose the high-price options with full-refund. The second branch captures the scenario wherein the honest returners self-select the no-returns-allowed option at price $p-\delta$. We next clarify two modeling choices we made in this section. First, note that we still permit $\hat{r}_R^t$ to take its optimal value for the optimization model that underlies expression (14). This is to maintain that the full-targeted-refunds proposal continues to function as a benchmark, i.e., an upper performance bound, for the full-menu-of-refunds proposal. Second, for the optimization model that underlies expression (15), we apply a 100% restocking fee for the customers who self-select the discounted price option. Even though this assumption does not exactly reflect the pricing model of Jet.com's, which permits returns at a cost to the customers, it strengthens our key managerial insight: If the full-menu-of-refunds proposal corresponding to (15) outperforms the full-targeted-refunds proposal corresponding to (14), a further optimized version of (15) would perform even better, possibly for an even wider range of problem scenarios. We provide complete characterizations of the three full-refunds proposal variations in Proposition **??** as well as the underlying pricing dynamics in subsection **??** within Appendix B. Here, the next proposition formalizes profit implications, which we illustrate in Figure 2.

PROPOSITION 7. *Denoting by*

$$\hat{X} \triangleq \frac{\big(V_L - \int_0^{V_L} G_L(v)dv - s\big) - \big(\hat{p}^u - s\big)\big(1 - G_L(\hat{p}^u)\big)}{\big(\hat{p}^t - \hat{r}_R^t\big)\big(1 - G_S(\hat{p}^t - \hat{r}_R^t)\big) + \big(V_L - \int_0^{V_L} G_L(v)dv - s\big) - \big(\hat{p}^u - s\big)\big(1 - G_L(\hat{p}^u)\big)}$$

a threshold value for renters' fraction in the customer base, for a given V_L, optimal profits $\hat{\Pi}^u$, $\hat{\Pi}^t$, and $\hat{\Pi}^m$ for the three full-refund proposals under consideration satisfy the following:

(a) $\hat{\Pi}^t > \hat{\Pi}^m = \hat{\Pi}^u$ *if* $s \geq \hat{s}$.

(b) $\hat{\Pi}^t > \hat{\Pi}^m > \hat{\Pi}^u$ *if* $s < \hat{s}$ *and* $X \geq \hat{X}$.

(c) $\hat{\Pi}^m > \hat{\Pi}^t > \hat{\Pi}^u$ *if* $s < \hat{s}$ *and* $X < \hat{X}$.

Evident in Figure 2 is that Proposition 7(a) applies to scenarios with salvage values exceeding $\hat{s}$ irrespective of how large renters' fraction X is. For these high-clearance-price scenarios, not only does full-targeted-refunds outperform the other two, but full-menu-of-refunds fails to segment customers. In contrast, parts (b) and (c) highlight that full-menu-of-refunds segments consumers when the salvage value is below $\hat{s}$, thus outperforming full-uniform-refunds. Nevertheless, whether segmentation by full-menu-of-refunds outperforms full-targeted-refunds depends further on renters' fraction X. This result follows from relative magnitudes of each demand segment, and the corresponding margins the retailer enjoys from each segment under the two full-refund proposals. Specifically, full-menu-of-refunds suppresses honest returns by setting $\hat{p}_H^m - \hat{r}_H^m$ equal to $\hat{p}_H^m$, and thus enjoys margin $\hat{p}_H^m - c$ from both the *keep* and *return* segments, i.e., all honest returners, and

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loses $c-s$ from transactions with the *rent* segment. In other words, full-menu-of-refunds optimizes margins from honest returners to counter loses from renters, and thus works well only when the customer base has sufficiently large fraction of honest returners, i.e., $1-X > 1-\hat{X}$. On the other hand, full-targeted-refunds loses $c-s$ from the transactions with the *return* segment (of honest returners), while optimizing margins $\hat{p}^t - c$ and $\hat{p}^t - (c-s)$ from the *keep* and *rent* segments, respectively. As the latter margin from renters is evidently bigger than the former two margins from the honest returners, full-targeted-refunds outperforms full-menu-of-refunds when X exceeds $\hat{X}$ even if the salvage value s falls below $\hat{s}$.

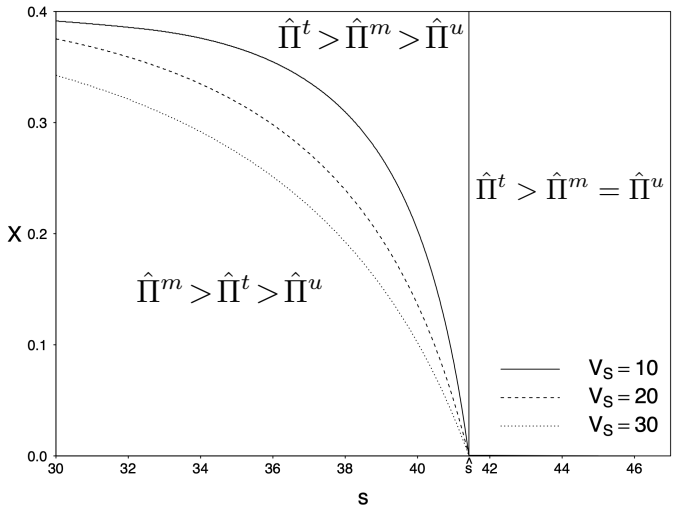


Figure 2 **Equilibrium profit comparisons for full-refund proposals as X and s change for three representative V_S values. Parameter values: $V_L = 100$, $c = 42$ and D follows Normal(100,20).**

Figure 2 further illustrates that threshold $\hat{X}$ decreases as $\frac{V_S}{V_L}$ increases, because it makes sense for the retailer to prioritize rental revenues (and avoid rent losses) for a larger set of problem scenarios as the rental-revenue-potential increases. This is promising for the applicability of full-menu-of-refunds in practice as one expects the renters' maximum willingness-to-pay V_S for short term use for most products to be significantly lower than the honest returners' maximum willingness-to-pay V_L for long term use. Also note in Figure 2 that even when $\frac{V_S}{V_L}$ approaches 1, i.e., its theoretical limit, the scenarios that support profit order $\hat{\Pi}^m > \hat{\Pi}^t$ still exist. This is promising for retailers who compete in business settings wherein full-refund practices are the norm, as implementing full-menu-of-refunds gives these retailers an edge over their competitors with technological (and financial) capabilities to track consumers necessary to implement full-targeted-refunds. Furthermore, even for these high $\frac{V_S}{V_L}$ scenarios, if the salvage value is sufficiently low, full-menu-of-refunds always achieves segmentation, and thus outperforming full-uniform-refunds, i.e., $\hat{\Pi}^m > \hat{\Pi}^u$ if $s < \hat{s}$.

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8. Alternative Model Specifications to Confirm Robust Findings

In this section, we extend the base model we formulated in §3 to conduct the analyses of three proposal alternatives in §§4, 5, and 6. Our goal here is to highlight how our primary managerial finding regarding the effectiveness of the menu-of-refunds proposal remains intact. Specifically, we consider the following alternative model specifications: In §8.1, we relax the assumption that the retailer recovers the same salvage value s on all returned products. In §8.2, we consider the possibility that some renters (with sufficiently high valuations for the product) may decide to keep the product beyond the originally intended short-term use. In §8.3, we relax the assumption that the targeted-refunds proposal identifies renters and honest returners correctly. In §8.4, we introduce *ex-ante* heterogeneity concerning honest returners' short-term as well as long-term valuations. Finally, in §8.5, we discuss implications of multi-period dynamics and transaction costs.

As we analyze each of these scenarios, we repeatedly leverage the demand dynamics that split uncertain aggregate demand D into *keep*, *return*, *rent*, and *no transaction* segments. Each relaxed base model assumption influences the ensuing demand segmentation, thereby affecting net marginal revenue M and fraction of transacting customers N. (Although M and N are functions of the retailer's price and refund decisions, we will present them without arguments for notational simplicity.) This approach also lets us make accurate comparisons relative to the base model formulation, and thus assess the impact of each relaxed assumption *ceteris paribus*.

We continue to use superscripts *"u"*, *"t"*, and *"m"* to distinguish uniform-refunds, targeted-refunds, and menu-of-refunds. Furthermore, we will utilize two-part acronyms *"xy"* to indicate optimal decisions and functions of optimal decisions for these proposals under relaxed assumptions. Specifically, x indicates proposal and is one of $\{u, t, m\}$, and *"y"* is one of $\{s, k, n, hs, hl\}$, where *"s"* denotes different **s**alvage values, *"k"* denotes renters may **k**eep products beyond initial use, *"n"* denotes targeted-refunds with **n**oisy identification, and *"hs"* and *"hl"*denote **h**eterogenous **s**hort-term and **h**eterogenous **l**ong-term valuations for honest returners, respectively. For example, r^{uk} denotes the optimal **u**niform-refunds when some renters **k**eep a rented product.

In the following subsections, we restrict our presentation to fundamental changes relevant to the retailer's optimization in each scenario, as well as the main finding concerning the efficiency of the menu-of-refunds proposal relative to the targeted-refunds proposal. We do so by leveraging Proposition 6, which we illustrate in Figure 1. This is because the conditions under which a retailer optimally implementing uniform-refunds by setting $r^u > r^b = s$ coincide with those under which a retailer optimally implementing targeted-refunds by setting $r^t_R > r^t_H$. In both cases, a retailer using menu-of-refunds with $r^m_R > r^m_H$ can segment customers as profitably as a retailer using

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targeted-refunds. (This is because $r_R^m > r_H^m$ implies $p_R^m - r_R^m < p_H^m - r_H^m$, i.e., price-refund pairs are incentive compatible.) When necessary, we provide further detailed technical analysis to support our conclusions for these alternative model specifications in Appendix B.

8.1. Different salvage values for returned products

It is conceivable that the salvage value for rented products s_R be different from the salvage value for those returned by honest returners s_H, as the anticipated wear-and-tear on each type of return might vary. We analyze this possibility by considering scenarios that support $s_H > s_R$ and $s_H < s_R$. The former scenario is when renters, who would certainly use the product, cause more wear-and-tear on the product as opposed to honest returners. This is because honest returners, upon realizing misfit, are unlikely to use the product at all. The latter scenario applies when the renters would be careful with their initial (and possibly only) use of the product, and return it early enough that the retailer could still resell without a steep discount. An honest returner, on the other hand, with the original intent/desire to keep the product may end up using the product several times, and return it later in the allowable returns period. In both scenarios, we set $s_H = s$ to permit comparisons with the base model we introduced in §3.

We start by noting the changes in how demand segmentation and margins change when s_H and s_R differ: (i) Sales to the *keep* segment would yield margin $p-c$, which we can re-write as $(p-r)+(r-s_H)-(c-s_H)$. (ii) Sales to the *return* segment would yield margin $(p-r)-(c-s_H)$. (iii) Transactions with the *rent* segment would yield margin $(p-r)-(c-s_R)$, which we can re-write as $(p-r)-(s_H-s_R)-(c-s_H)$. Finally, (iv) the *no transaction* segment would yield leftover inventory with negative margin $-(c-s_H)$. Then, we can relate the objective function of a retailer implementing uniform-refunds when $s_H \neq s_R$ to the same when $s_H = s_R$ as

$$\Big(M^u(p,r) - \big(s_H - s_R\big)X\big(1-G_S(p-r)\big)\Big)\mathbb{E}\min\left\{\frac{Q}{N(p,r)}, D\right\} - \big(c-s_H\big)Q, \tag{16}$$

and state the following proposition:

Proposition 8. *Denoting by r^{us} the value of r satisfying*

$$\left[\frac{\partial M^u(p,r)}{\partial N^u(p,r)}\, IU(Q,p,r) - \big(c-s_H\big)\right] - \big(s_H - s_R\big) = 0, \tag{17}$$

we have $r^{us} > r^u$ if-and-only-if $s_H < s_R$.

To appreciate the managerial relevance of Proposition 8, we leverage Proposition 6, which highlights that for scenarios supporting $r^u > s$ we have $\Pi^m = \Pi^t > \Pi^u$. Evidently, for these scenarios, we would have $r^{us} > r^u > s$, and thus $\Pi^{ms} = \Pi^{ts} > \Pi^{us}$, provided $s_H < s_R$. In other words, if the retailer can clear rented products at a higher price than honest returns, there would emerge more scenarios in which the menu-of-refunds proposal would be as effective as the targeted-refunds proposal.

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8.2. Possibility of some renters keeping the product

We now consider the possibility that some renters (with sufficiently high valuations for the product) may decide to keep the product beyond its short-term use. Obviously, if such renters exist, they would end up paying the full price p instead of the rental price $p-r$ when they opt to keep the product. In other words, whereas the renters whose valuations exceed p would keep the product, those with valuations between $p-r$ and p would continue to rent it. As before, renters with valuations below $p-r$ would find it too expensive to even rent the product, and choose not to transact. The ensuing demand segmentation and margins would be as follows: (i) The *keep* segment would expand to fraction $(1-X)\,(1-G_L(r))+X\,(1-G_S(p))$ of the demand (relative to when no renters keep) and yield margin $p-c$, which we can re-write as $(p-r)+(r-s)-(c-s)$. (ii) Sales to the *return* segment would yield margin $(p-r)-(c-s)$. (iii) Transactions with the *rent* segment would contract to $X\,(G_S(p)-G_S(p-r))$ of the demand (relative to when no renters keep) and yield margin $(p-r)-(c-s)$, which we can re-write as $(p-r)-(c-s)$. Finally, (iv) the *no transaction* segment would yield leftover inventory with negative margin $-(c-s)$. Then, we can relate the profit of a retailer implementing uniform-refunds when all renters return to the same when renters with high valuations keep as in the following

$$\Big(M^u(p,r)+\big(r-s\big)X\big(1-G_S(p)\big)\Big)\mathbb{E}\min\left\{\frac{Q}{N^u(p,r)},D\right\}-\big(c-s\big)Q. \tag{18}$$

The solution to this optimization problem yields the following proposition:

Proposition 9. *Denoting by r^{uk} the value of r satisfying FOC*

$$\Big[\frac{\partial M^u(p,r)}{\partial N^u(p,r)}\,IU(Q,p,r)-\big(c-s\big)\Big]+\frac{\partial\big[X(r-s)(1-G_S(p))\big]/\partial r}{\partial N^u(p,r)/\partial r}=0. \tag{19}$$

we have $r^{uk}<r^u$ if $r^u<s$, in which case $\Pi^{mk}<\Pi^{tk}$.

As was the case concerning the interpretation of Proposition 8, the relative magnitudes of r^{uk} and r^u are again important to interpret Proposition 9, as condition $r^u>s$ is sufficient to ensure that the menu-of-refunds proposal successfully distinguishes renters from honest returners. Evidently, the propensity of some renters turning into customers who end up keeping the product restricts the range of scenarios for which menu-of-refunds proposal overperforms the targeted-refunds proposal. This is because, in addition to all scenarios where $r^u<s$, even when $r^u>s$ it is possible that we have $r^{uk}<s<r^u$. Therefore, if the retailer learns that it can convert renters to customers who keep the product, it should resort more to the targeted-refunds proposal to ensure that it properly optimizes the rental fee, and thus the rental revenues.

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8.3. Targeted-refunds proposal with noisy identification

We now relax the assumption that the targeted-refunds proposal always identifies renters and honest returners correctly. Specifically, we assume that the retailer implementing targeted-refunds is still aware that fraction X of the aggregate customer base is renters; however, it incorrectly offers r_H (r_R) to a renter (honest returner) with probability γ_R (γ_H). In other words, the retailer may forecast fraction X (for example, based on NRF annual survey results, or its own transactional data), but correctly identifying each renter is a separate capability related to the accuracy of the third-party software solution being implemented, the accuracy of which we model by probabilities γ_R and γ_H. As a result, both N^t and M^t are revised as follows:

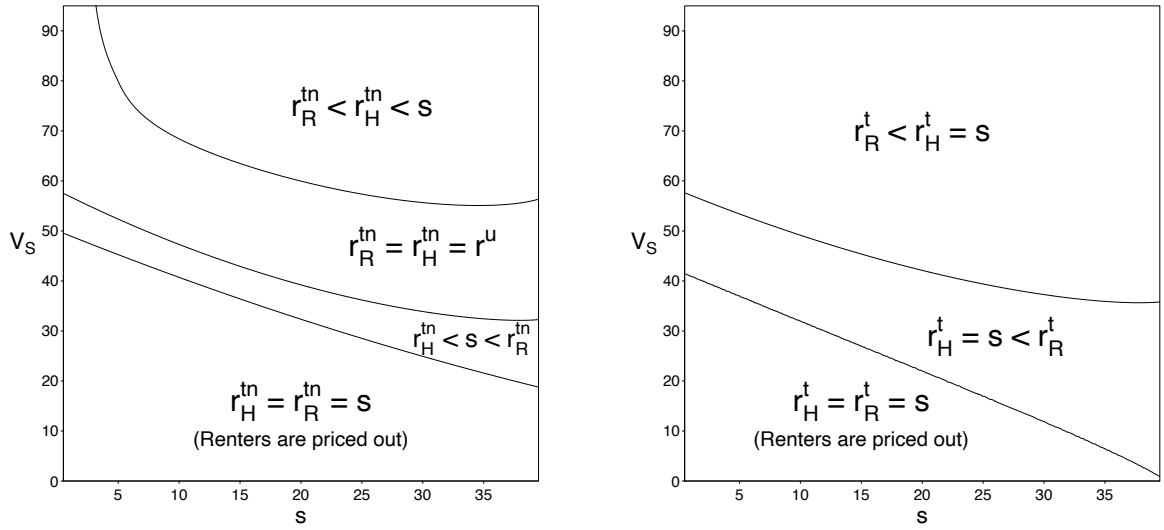
$$N^{tn} = N^t - \begin{cases} X\gamma_R\big(G_S(p-r_H) - G_S(p-r_R)\big) & \text{if } r_R \geq r_H \\ \Big(X\gamma_R\big(G_S(p-r_H) - G_S(p-r_R) + (1-X)\gamma_H\big)\Big) & \text{if } r_R < r_H \end{cases} \tag{20}$$

$$\begin{aligned} M^{tn} = M^t &- X\gamma_R\big(\big(p-r_R\big)\big(1-G_S(p-r_R)\big) - \big(p-r_H\big)\big(1-G_S(p-r_H)\big)\big) \\ &- \begin{cases} \big(1-X\big)\gamma_H\big((r_R-s)G_L(r_R) - (r_H-s)G_L(r_H)\big) & \text{if } r_R \geq r_H \\ \big(1-X\big)\gamma_H\big((p-s) - (r_H-s)G_L(r_H)\big) & \text{if } r_R < r_H \end{cases} \end{aligned} \tag{21}$$

Revised values N^{tn} and M^{tn} reflect the impact of misidentification depending on the relative magnitudes of r_R and r_H. Concerning the renters who are misidentified as honest returners, which represent $X\gamma_R$ of the customer base, expression (20) highlights that these customers, facing refund r_H, would transact only if their short-term valuations exceed the ensuing rental fee $p-r_H$. When $r_R > r_H$, this would decrease N^t, whereas when $r_R < r_H$ more renters would transact. Similarly, revised net margin expression (21) reflects the fact that the retailer would enjoy rental fee $p-r_H$ instead of $p-r_R$ on this segment. Concerning the honest returners misidentified as renters, who represent $(1-X)\gamma_H$ of the customer base, expression (20) highlights that these customers still buy when $r_R \geq r_H$, but they walk without a purchase when $r_R < r_H$. This is because the retailer sets price $p = \mathbb{E}\max\{u_L, r_H\}$, yielding a positive surplus $\mathbb{E}\max\{u_L, r_R\} - p > 0$ when $r_R > r_H$. Consequently, expression (21) reflects both the revised return rate $G_L(r_R)$ instead of $G_L(r_H)$ as well as the revised contribution $r_R - s$ instead of $r_H - s$ to the net margin. When $r_R < r_H$, misidentified honest returners do not transact as $\mathbb{E}\max\{u_L, r_R\} - p < 0$, and thus revised net margin expression (21) reflects only the lack of contribution $\big((p-s) - (r_H-s)G_L(r_H)\big)$ from this segment.

We provide detailed analysis of the targeted-refunds proposal with noisy identification in subsection **??** within Appendix B, where Proposition **??** fully characterizes the retailer's optimal decisions. We illustrate these optimal decisions in Figure 3, wherein panel (a) shows the impact of noisy identification relative to panel (b) where targeted-refunds achieves 100% accuracy. Figure 3(a) shows that noisy identification forces the retailer to set $r_R^{tn} = r_H^{tn} = r^u = s$ for scenarios with intermediate V_S values. In other words, the retailer ignores noisy identification and chooses not to discriminate. For these scenarios, as Figure 3(b) illustrates, noise-free targeted refunds would have

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(a) Targeted-refunds with $\gamma_H = 0.02$ and $\gamma_R = 0.02$. **(b)** Targeted-refunds with $\gamma_H = \gamma_R = 0$

Figure 3 Implications of targeted-refunds with noisy identification.

set $r_R^t > s = r_H^t$. Similarly, for the same scenarios, menu-of-refunds would have set $r_R^m > s = r_H^m$ and segmented customers, and thus over-performing relative to targeted-refunds with noisy identification. Evidently, as γ_H and/or γ_R increase, this inefficiency due to noisy identification increasingly restricts the scenarios in which targeted-refunds can outperform menu-of-refunds to only those with high V_S and high s, i.e., scenarios marked $r_R^{tn} < r_H^{tn} < s$. Finally, Figure 3(a) also shows that targeted-refunds with noisy identification yields more scenarios wherein renters are priced out, an outcome menu-of-refunds would have achieved without any implementation costs.

8.4. Heterogenous valuations for honest returners

In this subsection we introduce two layers of heterogeneity for honest returners' valuations of the product to highlight that the main finding concerning the efficacy of the menu-of-refunds proposal we obtained using our base model holds without loss of generality. Recall that the base model we introduced in §3 assumes $u_H = u_L$, which implies that, in contrast with renters, honest returners do not associate any known short-term valuation with the product, i.e., $u_S = 0$. Furthermore, we have so far assumed that unknown long-term valuation u_L for each honest returner is drawn from the same probability distribution G_L; in other words, honest returners may have heterogenous valuations *ex post* but are homogenous *ex ante*. Relaxing each of these assumptions would yield scenarios wherein it is possible that some honest returners not buy the product.

8.4.1. Heterogenous short-term valuations for honest returners. We now let honest returners have customer-specific short-term utility u_S^H in addition to their long-term utility u_L. We assume u_S^H follows distribution $G_S^H(\cdot)$ with support over $[0, V_S^H]$. We further allow renters' short-term utility to differ from honest returners' short-term valuation, and denote it by u_S^R which follows

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distribution $G_S^R(\cdot)$ with support over $[0, V_S^R]$. As such, the total utility of a honest returner from the product is $u_H = u_L + u_S^H$, and she makes a purchase decision by comparing $u_S^H + \mathbb{E}\max\{u_L, r\}$ and price p. In other words, only the honest returners whose short-term valuation u_S^H exceeds $p - \mathbb{E}\max\{u_L, r\}$ purchase. Note that the short-term utility is irrelevant for the return decision of honest returners, and a fraction $G_L(r)$ of honest returners who purchase the product, return it upon learning u_L. This demand model yields the following segmentation: (i) The *keep* segment corresponds to fraction $(1-X)(1-G_L(r))(1-G_S^H(p-\mathbb{E}\max\{u_L, r\}))$; (ii) the *return* segment represents fraction $(1-X)G_L(r)(1-G_S^H(p-\mathbb{E}\max\{u_L, r\}))$; (iii) the *rent* segment represents fraction $X(1-G_S^R(p-r))$; and, *no transaction* segment represents fraction $XG_S^R(p-r) + (1-X)(G_S^H(p - \mathbb{E}\max\{u_L, r\}))$. Note that as compared to our main model, the last segment grows at the expense of reduction in the first two segments. As a result of these changes, functions N and M representing fraction of customers intending to purchase the product and net unit revenue, respectively, change under each refund proposal, leading to modified expressions for profit functions as well as modified characterization of optimal decisions. We present detailed formulations and complete characterizations of the optimal decisions for the three proposals subject to the aforementioned demand model in subsection **??** within Appendix B, and highlight the most significant managerial implication of our analysis in the next proposition:

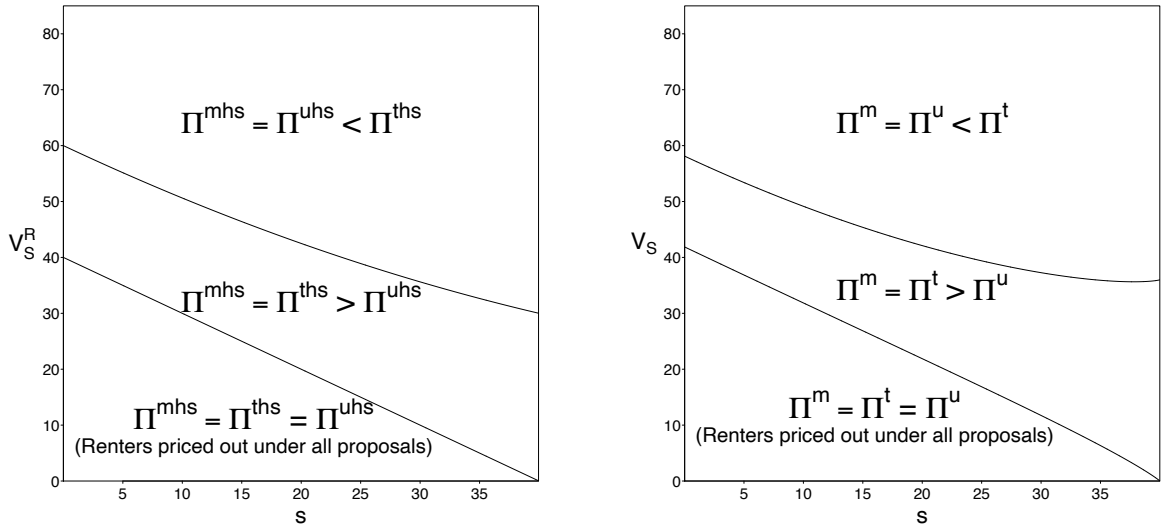
PROPOSITION 10. *There exists a threshold value of V_S^R, such that for values of V_S^R larger than this threshold value $\Pi^{ths} > \Pi^{mhs} = \Pi^{uhs}$, and $\Pi^{ths} = \Pi^{mhs} \geq \Pi^{uhs}$, otherwise.*

Figure 4 complements Proposition 10 by illustrating the profit implications of honest returners' heterogenous short-term valuations. Most strikingly, menu-of-refunds remains as effective as targeted-refunds when $V_S^H > 0$ in all low-rental-value-potential scenarios just as is the case when $V_S^H = 0$. The range of such scenarios for which $\Pi^{ths} > \Pi^{mhs} = \Pi^{uhs}$ expands as V_S^H increases, confirming that the primary finding of our base model formulation sustains without loss of generality.

8.4.2. Heterogenous long-term valuations for honest returners. To model honest returners' heterogenous long-term valuations, we assume fraction α are *high-type*, and the rest are *low-type*, and denote type-specific quantities by superscripts "h" and "l," respectively. We let these valuations satisfy $u_H^i = u_L^i$ for $i = \{h, l\}$ where u_L^i is drawn from G_L^i with support $[0, V_L^i]$. We further assume that G_L^h stochastically dominates G_L^l, i.e., $G_L^h(x) < G_L^l(x)$ for all x, and $V_L^h \geq V_L^l$. Lastly, we set $G_L^l = G_L$ to permit comparisons relative to the base model we introduced in §3.

We first note the necessity to consider two possible sets of price and refund values: (i) Setting $p = \mathbb{E}\max\{u_L^l, r\}$ to ensure all honest returners buy, *or* (ii) setting $p = \mathbb{E}\max\{u_L^h, r\}$ to price out

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(a) $V_S^H = 20$ and $V_L^H = 80$ **(b)** $V_S^H = 0$ and $V_L^H = 100$

Figure 4 Equilibrium profits for each proposal when honest returners enjoy short-term utility, i.e., $V_S^H > 0$, versus when they do not, i.e., $V_S^H = 0$. Parameter values: $V_L^H + V_S^H = 100$, $c = 40$ and D follows Normal(100,20).

low-type honest renters. In the first scenario, as $G_L^l = G_L$, total transactions with honest returners when $\alpha > 0$ remains the same as when $\alpha = 0$. Nevertheless, as G_L^h stochastically dominates G_L^h, returns volume for any r by honest returners when $\alpha > 0$ is lower than the same when $\alpha = 0$, i.e., $\alpha G_L^h(r) + (1-\alpha)G_L^l(r) < G_L^l(r) = G_L(r)$. As such, the retailer can set $r^{uhl} > r^u$, which in turn yields $p^{uhl} > p^u$ for scenario (i). This naturally extends to scenario (ii), the price and the refund for which, by definition, exceed those for the scenario (i). The next proposition highlights the managerially consequential findings that follow from these aforementioned dynamics:

PROPOSITION 11. *For all scenarios supporting $r^u > r^b = s$, and thus $\Pi^m = \Pi^t > \Pi^u$, we have $r^{uhl} > r^u > r^b = s$, and thus $\Pi^{mhl} = \Pi^{thl} > \Pi^{uhl}$.*

Proposition 11 leverages the fact that $r^{uhl} > r^u$ implies $r^{uhl} > r^b = s$ for all scenarios supporting $r^u > r^b = s$. As the latter conditions also implies $\Pi^m = \Pi^t > \Pi^u$, we can conclude that menu-of-refunds is as effective as targeted-refunds in even more scenarios when $\alpha >$ relative to when $\alpha = 0$. Finally, we can relax assumption $G_L^l = G_L$ to attain the following corollary:

COROLLARY 1. *Proposition 11 holds as long as $\alpha G_L^h(r^{uhl}) + (1-\alpha)G_L^l(r^{uhl}) < G_L(r^u)$.*

In words, *ex-ante* heterogeneity of honest returners' long-term valuations increases the range of scenarios for which menu-of-refunds is effective as targeted-refunds as long as uniform-refunds with r^{uhl} generates fewer returns by honest returners than r^u.

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8.5. Implications for multiple period settings and transaction costs

Our two-period models with exogenous salvage revenues do not address managing opportunistic returns over multiple periods. Nevertheless, several model features embedded in our base formulation as well as the alternative model specifications we considered allow us to offer *limited* insights concerning the potential effectiveness of menu-of-refunds in multiple period settings with static decisions, i.e., the one wherein the retailer sets Q, p, r before the initial period. More specifically, we can interpret stochastic demand D in our formulation as total demand realized across several periods, which the retailer would try to satisfy by stocking Q at the beginning of the planning horizon. The retailer following a markdown pricing scheme would sell this product at price p in earlier periods while offering r for returns, re-sell early returns (i.e., possibly by renters) and remaining inventory at s_R during middle periods, and sell the rest of its inventory as well as later returns (possibly by honest returners) at s_H, which may be lower than c. Fraction $1-N$ of customers would find the prices they face prohibitively high and not transact, thus making up the *no transaction segment*. The remaining fraction N of customers would split into the *kept sales*, *return*, and *rent segments*, generating different margins based on the prices they face during the period they transact. As these essential demand dynamics remain intact even when there is further heterogeneity in customers' purchase and return behavior and targeted-refunds suffers from noisy identification, menu-of-refunds proposal *may remain* a viable strategy to combat opportunistic returns. Admittedly, characterizing the optimal dynamic pricing and inventory decisions is beyond the scope of this paper. Nevertheless, considering the retailer's multiple replenishment opportunities as well as consumers' repeated interactions and changing renting behavior over multiple periods are worthy research directions.

Our analysis also assumes frictionless returns for consumers, i.e., denoting the transaction cost of making a return by t, we assume $t=0$. In reality, while this may be the case for renters, who are predisposed to return the product, for honest returners, it is possible that $t>0$. Altug and Aydinliyim (2016) modeled this possibility by expressing *ex-ante* consumer surplus as $\mathbb{E}\{r-t\,,u_L\}-p$, and showed a retailer's response to $t>0$ is nuanced. Specifically, a retailer implementing uniform-refunds may lower r^u *or* increase it to p^u, particularly if t is sufficiently high. This finding has the following implication for this paper: When it is optimal for the retailer to increase r relative to when $t=0$, we expect menu-of-refunds to be as effective as targeted-refunds in more scenarios.

9. Discussion and Conclusion

As the retail industry has gone through a significant revitalization over the last two decades, it became paramount for retailers to find ways to stimulate demand, and enhance customer experience

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not only during the actual shopping process, but also in the aftermath of a purchase. One particular tactic retailers have increasingly used is to give procrastinating customers ease of mind by offering lenient return policies. By doing so, retailers effectively provide a low cost (or, sometimes free) trial option, thereby reducing regret due to potential misfit between a product and prior customer expectations of it. However, honoring this after-sales service commitment is a costly proposition for retailers for several reasons, including the obvious reverse logistics costs, and the necessity to mark down some returned products such as open-box electronics. These lenient return policies may also yield unintended consequences such as opportunistic consumer returns by customers who purchase a product without the intent to keep it beyond its short-term (oftentimes single) use. Assessing the effectiveness of strategies retailers implement to mitigate the adverse consequences of such renting behavior is the primary objective of this study.

We particularly focused on two proposals, namely targeted-refunds and menu-of-refunds, and compared and contrasted the optimal decisions and profit implications of both proposals with respect to two benchmark settings. The first benchmark setting is the one Su (2009) studied with no renters among the retailer's customers, and the second one is one in which the retailer implements uniform-refunds, where the retailer optimizes its decisions while taking into account the emergence of renters. Our motivation for considering targeted-refunds is the emergence of third-party retail analytics firms such as The Retail Equation (TRE), which has increasingly partnered with major retailers. These "return authorization" software solutions TRE provides processes customers' past purchase and return transactions to predict whether s/he could be a renter. Then, the retailer targets such customers by offering them a less lenient return option should the customers go through with the purchase despite that restriction. We also considered menu-of-refunds several online retailers such as Jet.com use to segment their customers based on their willingness to maintain a low cost return option. This policy, by presenting customers with multiple price-refund pairs, can help distinguish honest returners, who would return only if they are unsatisfied with their purchase, from renters who are pre-disposed to return after initial use.

Our findings make a case for both business models, i.e., targeted-refunds and menu-of-refunds,, while issuing caution to retail managers that the market, cost, and price dynamics that support the optimality of each proposal are distinct. We first analyzed the case where the retailer can apply restocking fees for all returns, the findings for which we summarize in Figure 5. In this case, a retailer implementing targeted-refunds can correctly distinguish renters from honest returners, and thus cannot do worse than when implementing other alternatives. We found that menu-of-refunds can match the optimal profit of targeted-refunds in a market with low-rental-value potential

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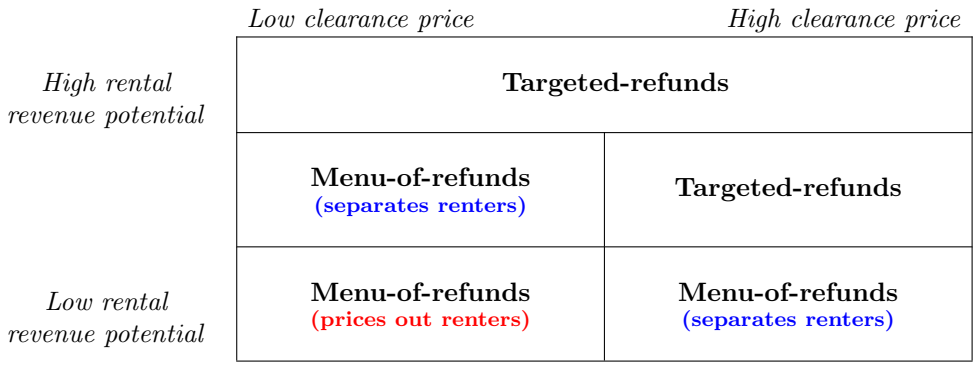


Figure 5 **Optimal proposal and underlying mechanism for why it works. For scenarios marked "Targeted-refunds," menu-of-refunds fails to separate customer types, and thus is outperformed.**

or when the retailer must liquidate leftover units and returns at a low clearance price even in a medium-rental-revenue-potential market. We showed that this proposal successfully segments customers by directing renters to the high-price-high-refund pair with its lower rental fee relative to the restocking fee of the low-price-low-refund pair.

9.1. Managerial implications

Our findings for the aforementioned business models with optimal restocking fees have significant managerial consequences, as they contradict the seemingly evident pricing reaction of a retailer to emerging renters among its customers, which would be to implement a more strict return policy by increasing restocking fees. Not only do we show that the optimal response should be to lower rental fees for renters, but also we highlight why this is the right thing to do by shedding light on demand segmentation dynamics that underlie the successful pricing scheme of notable implementers of menu-of-refunds in practice such as Jet.com. Furthermore, the optimal price-refund pairs we derive are incentive compatible, and reflect the realistic pricing practices of retailers that successfully implement this proposal. Specifically, at equilibrium, we find the price-differential of menu-of-refunds to be less than its refund-differential. This pricing structure happens to coincide with Jet.com's pricing principles where the customers who choose the low-price option with 2-3% discount are subjected to a 5% restocking fee plus a flat fee of $5.99 should they return their purchases as opposed to the no-restocking-fee return option customers who pay the full price can enjoy. This pricing scheme is also consistent with the findings of a recent study by Shang et al. (2017b), which prices free returns at about 3% of product value using evidence from eBay.

Our analysis of several alternative model specifications highlight that the main managerial insight our base model produced regarding the effectiveness of menu-of-refunds is robust. In Figure 6, we highlight that several modeling assumptions we relax for these alternative models make our result

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stronger. For example, a higher salvage value on rented product than on honest returns as well as introducing *ex-ante* heterogeneity in honest returners valuations expand the range of scenarios for which menu-of-refunds is as effective as targeted-refunds. On the other hand, the more likely it is that some renters keep the products they purchase, the more likely it becomes that targeted-refunds outperforms menu-of-refunds.

Extension	**Requisite condition(s) for the conclusion**	**Conclusion regarding the relative performance of the proposals**
Full-refund proposal variations **Subsection 7.2**	Low (high) clearance price and (or) few (many) renters	Menu-of-refunds performs better (worse) than targeted-refunds.
Different clearance price for renters and honest returners **Subsection 8.1**	Higher (lower) clearance price for earlier returns	Menu-of-refunds outperforms targeted-refunds in more (fewer) cases.
Some renters may keep the product **Subsection 8.2**	Always	Targeted-refunds outperforms menu-of-refunds in more cases.
Targeted-refunds with noisy identification **Subsection 8.3**	Always	Menu-of-refunds outperforms targeted-refunds in more cases.
Heterogenous short-term valuations for honest returners **Subsection 8.4.1**	Always	Menu-of-refunds outperforms targeted-refunds in more cases.
Heterogenous long-term valuations for honest returners **Subsection 8.4.2**	Lower (higher) returns volume	Menu-of-refunds outperforms targeted-refunds in more (fewer) cases.

Figure 6 Relative performance of menu-of-refunds and targeted-refunds in extensions considered.

Our findings also make a business case for using software solutions such as the one provided by TRE under certain conditions. Specifically, we argue that retailers operating in high-rental-revenue-potential markets should invest in return authorization software solutions, as menu-of-refunds cannot segment customers in such scenarios. Furthermore, rental fees for these returns should be optimized as opposed to being set at prohibitively high values. Nevertheless, our analysis of targeted-refunds with noisy identification issues caution for the potential adopters of these software solutions. In particular, we find that identification errors limit the effectiveness of targeted-refunds *only* to high-clearance-price-products sold in significantly-high-rental-revenue-potential markets. In contrast, for low-clearance-price-products sold in medium-rental-revenue-potential markets, we find that noisy targeted-refunds would price out renters more aggressively than it should, and thus be outperformed by menu-of-refunds, which would optimize rental revenues from renters.

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Reflecting the common use of "no-questions-asked" return policies several retailers implement, we also assess the three proposals under consideration with an additional provision that the retailer would waive its restocking fees on at least one type of customer returns. Evidently, the full refund provision applies to all returns if the retailer is to implement the full-uniform-refunds proposal. For the full-targeted-refunds proposals we assume the retailer offers the full refund provision only to the honest returners and optimize the refund it offers to renters. (In most practical applications, the renters' refund is set at 0.) For the full-menu-of-refunds proposal, we let the retailer to offer 100% refund only to customers who self-select the high-price option, while setting the refund to 0 for the low-price option. (In most practical applications, the low-price options still include positive refunds.) In other words, we purposely boost the capability of the full-targeted-refunds proposal while handicapping the full-menu-of-refunds proposal. Nevertheless, we find the full-menu-of-refunds proposal to strictly outperform the full-targeted-refunds proposal when the salvage value is below a threshold and the renters' population is sufficiently limited. It is of note that both of these necessary conditions for the effective use of this self-segmentation pricing strategy are realistic, as opportunistic returns account for only 11% of all returns according to the 2017 NRF survey, and are prevalent particularly for returned products that must be discounted severely such as open-box electronics, or fashion products for which returns that take place towards the end of short selling seasons are usually cleared at a loss. Our study shows that it is possible for such retailers to combat the adverse profit consequences of opportunistic consumer returns by implementing innovative-yet-simple price-refund schemes, which still permit full-refund returns, and remain even more effective than alternative targeted-refunds solutions requiring upfront and implementation costs.

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